Reg. No. | | | | | | | | | | |

[12ME836]

EIGHTH SEMESTER B.E. EXAMINATION, JUNE - 2016

NON TRADITIONAL MACHINING

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Write neat sketches wherever necessary.

1.	a.	Classify non traditional machining process based on the type of energy used and explain with an example.	07
	b.	Compare the traditional and non traditional machining processes.	07
2.	a.	Explain ultrasonic sinking and ultrasonic contour machining with a sketch.	10
	b.	List the factors affecting ultrasonic machining process.	04
3.	a.	Mention and explain the factors (variables) affecting material removal rate in Abrasive jet machining (AJM)	07
	b.	Sketch and explain EDM process.	07
4.	a.	Briefly explain the steps involved in chemical blanking.	08
	b.	List the etchants and maskents used for chemical milling.	06
5.	a.	Explain Electrochemical deburring.	05
	b.	Explain the mechanism of MRR in plasma arc machining (PAM). How do you differentiate PAM and plasma arc cutting (PAC)	09
6.	a.	Explain laser beam machining with a neat sketch.	08
	b.	List the types of lasers and commonly used gases in laser.	06
7.	a.	Differentiate non traditional and high energy forming process.	05
	b.	Classify non conventional methods of sheet forming. Explain any one with sketch.	09
8.		Write short notes on the following: (2×7=14)	14

i. Abrasive Slurry
ii. EDM Wire cutting
iii. MRR
iv. Ultrasonic frequency
v. Controlled machining
vi. Hydraulic forming
vii. Grit Size